DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2012.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE of the following questions.

1. Convert the following

   (a) $11001.101_{10} \Rightarrow (?)_8 \Rightarrow (?)_{16}$

   (b) $5073.625_8 \Rightarrow (?)_2 \Rightarrow (?)_{10}$

2. Discuss the basic laws of Boolean Algebra.

3. (a) With examples discuss on sum of products and prodcut of sums.

   (b) With an example discuss Karnaugh map simplification.

4. What is an half adder? Design a half adder and implement it using Universal gates.

5. What is a sequential circuit? Discuss with an example.

6. What do you mean by excitation table? Discuss the excitation table of a JK flip –flop.

7. What do you mean by primary and secondary memory?

8. Write a short note on computer generations.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. With examples discuss on the following :

   (a) Alpha-numeric codes

   (b) Error detection codes.

10. With Quine Mckluskey method simplify the following :

    $f(a, b, c, d) = \sum(0,1,3,4,7,9,11,12) + \sum(2, 8, 13, 15)$

    Verify the results using Karnaugh map.

11. With necessary diagrams discuss on the following :

    (a) Decoder

    (b) Multiplexer.

12. Write a short note on Binary parallel adder.

13. With necessary diagrams discuss the function of an arithmetic circuit.

14. Discuss on different Input-Output Units.

15. With necessary diagrams discuss the function of a BCD counter.

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 year Integrated) DEGREE EXAMINATION, MAY 2012.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Give the character set of C. List the features of C.

2. Explain the qualifiers with examples.

3. What are functions? Discuss the different type of functions.

4. What is meant by call by value? Explain parameter passing with an example.

5. What are pointers? Discuss the advantages of pointers.

6. What is a file? Explain how opening and closing a file are carried out in C.

7. What is Data structure? Discuss the structure of stack.

8. What are the advantages of Linked lists? How to delete or insert a node in a linked list?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) List the different data types supported by C. Discuss with examples.

   (b) With examples explain the various logical and relational operators in C.

10. What are the control structures available in C? Explain them with examples.

11. (a) Discuss on the following:

   (i) Macros

   (ii) C preprocessor.

   (b) With suitable examples, explain recursive functions.

12. (a) Write C program to convert a two dimensional array to one-dimensional array.

   (b) Write a C program that inverts a given string.

13. Discuss on the following:

   (a) Dynamic memory allocation

   (b) Structure.

14. Compare Queue and stack. List the applications of stack. Discuss any two applications of stack.

15. (a) What is Binary tree? How a List can be represented in a Binary tree?

   (b) How to insert an element in a doubly linked list? Discuss.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2012.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Write the following statement in symbolic form : If either Ravi takes French or Raja takes Tamil, then Raju will take Malayalam.

   (b) Construct the truth table for $Q \wedge (P \to Q) \to P$. (3 + 5)

2. Check whether the following formula is a Tautology or a contradiction. $\neg(P \vee (Q \wedge R)) \rightleftarrows (P \vee Q) \wedge (P \vee R)$.

3. (a) If $A = \{X, Y, Z\}$ write the power set of $A, P(A)$.

   (b) If $A = \{2,5,6\}, B = \{3,4,2\}$, find $A - B$ and $B - A$. (4 + 4)

4. (a) Define a n-ary relation.

   (b) Give a relation which is both a partial order relation and an equivalence relation on a set. (3 + 5)

5. Write any two functions from $X = \{a, b, c\}$ to $Y = \{0,1\}$ and indicate in each case whether the function is one-to-one, is onto and is one-to-one onto.

6. Define a semigroup and a monoid. If N is the set of natural numbers, prove that $< N, + >$ and $< N, \times >$ are monoids.

7. Prove that the set of all permutations on $G=\{a,b,c\}$ is a group under composition of functions.

8. (a) Define a bipartite graph and prove that every tree is a bipartite graph.

   (b) State true or false and justify: Every bipartite graph is a tree. (4 + 4)

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Show that $P \to (Q \to R) \Leftrightarrow P \to (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \to R$.

10. (a) Show that for any two sets A and B, $A-(A \cap B)=A-B$.

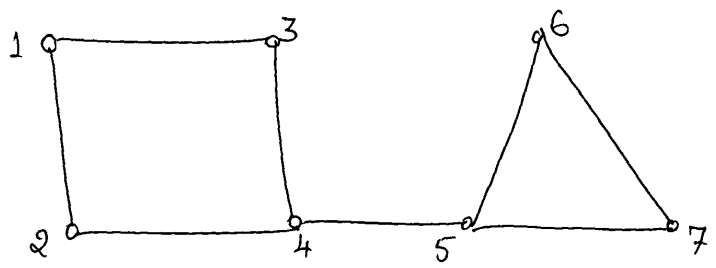
    (b) Let $X=\{1,2,\cdots,7\}$ and $R\{<x,y>/x-y \text{ is divisible by } 3\}$. Show that R is an equivalence relation. (7 + 8)

11. State and prove Lagrange's theorem.

12. Define a normal subgroup of a group G. Show that if G is an abelian group, then every subgroup of G is normal.

13. (a) Define a planar graph. Give an example.

    (b) Write the adjacency matrix and incidence matrix of the graph given below. (5 + 10)



14. (a) Define recursively defined functions and explain with examples.

(b) Write an algorithm that determines whether a given integer is a prime number. Discuss the complexity of the algorithm. (6 + 9)

15. Give a brief account of each of the following :

(a) Konigsberg bridge problem

(b) Traversable multigraphs

(c) Weighted graphs. (5 + 5 + 5)

DE–4213 **23**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (Five years Integrated) DEGREE EXAMINATION, MAY 2012.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum :100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What is objected oriented programming? List the advantages of object oriented programming.

2. What are the applications of C++? Describe the major parts of a C++ programming.

3. What is a class? Discuss with an example. How does a C++ structure differ from a C++ class?

4. What is a friend function? Discuss the characteristics of a friend function.

5. What is a pointer? With examples explain the arithmetic operations on pointers.

6. With examples discuss on the following :

   (a) Polymorphism

   (b) Operator overloading.

7. What is Inheritance? Describe the syntax of the single inheritance.

8. What do you mean by static and dynamic binding? Discuss with examples.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the basic concepts of object-oriented programming. List the applications of object-oriented programming.

10. (a) Discuss on C++ keywords.

    (b) Discuss on the size and range of C++ data types.

11. (a) What is a constructor? Discuss some of the properties of the constructor functions.

    (b) Write a C++ program to discuss the use of overloaded constructors.

12. (a) With a programming example discuss the use of pointers to objects.

    (b) What does the pointer do? Discuss the applications of this pointer.

13. (a) List the important functions supported by the string class.

    (b) What do you mean by overloading of a function? Discuss with example.

14. With a program illustrate how constructors are implemented when the classes are inherited.

15. Discuss on the following :

    (a) Virtual function

    (b) Derived and base classes.

DE–4214 **24**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 year Integrated)
DEGREE EXAMINATION, MAY 2012.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What is an operating system? Discuss the different types of services of OS.

2. What do you mean by system call? List the type of system calls.

3. What is process management? Discuss.

4. Write a short note on interprocess communication.

5. What is deadlock? How deadlock can be avoided?

6. What is paging? Discuss any one page replacement algorithms.

7. Write a short note on Buddy system.

8. What is a file? Discuss the operation of files.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. (a) Discuss on the history of operating systems.

   (b) What is a time-sharing? Discuss the differences between real-time operating system and time-sharing operating system.

10. What is semaphore? Discuss any two applications of semaphore.

11. What is Process scheduling? List the different process scheduling techniques. Discuss any three in detail.

12. (a) What is swapping? Discuss.

    (b) Write a short note on multiprogramming with fixed partitions.

13. What is disk scheduling? Discuss any three Disk space management.

14. Write a short note on the following :

    (a) Single-level directory

    (b) Tree-structured directory.

15. Discuss on the following :

    (a) File servers

    (b) Protection mechanisms.

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DE–4215 **25**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (Five year Integrated) DEGREE EXAMINATION, MAY 2012.

RELATIONAL DATABASE MANAGEMENT SYSTEMS

Time : Three hours Maximum :100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Write detailed notes on schema and subschema.

2. Explain in detail the various user interface with relevant examples.

3. Illustrate the basic concepts of relational model with suitable example.

4. Write an Overview about Relational Model.

5. Describe the Oracle Database Architecture.

6. Discuss about various types of databases with an example.

7. How to join multiple tables and how to select distinct values? Explain with an example.

8. Write notes on the following :

(a) Copying a Table

(b) Dropping a Table

(c) Modifying a Table

(d) Renaming a Table.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an Overview about features and advantages of Database Management System.

10. Explain the following :

(a) Data Manipulation facilities (7)

(b) Data Definition facilities. (8)

11. Discuss about Relational Database Management system.

12. Describe the Data access SQL Commands and DML Commands.

13. Write detailed notes on sub queries and multiple part queries.

14. Give detailed description about Embedded SQL and its statements.

15. Create a Table for your department and add all privileges required for the table.

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DE–4413 **16**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2012.

C PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

One question to be selected by lot system.

Each sub-division carries 50 marks.

1. (a) Write a C program to find factorial of a number using non-recurive method.

   (b) Write a C program to print reversed pyramid of digits.

---------- Cut here ----------

2. (a) Write a C program to find the prime numbers between two numbers entered by the user.

   (b) Write a C program to find the roots of a quadratic equation.

---------- Cut here ----------

3. (a) Write a C program using functions to replace a substring in a string and to find the length of a string.

   (b) Write a C program to find the largest and smallest element in an array

4. (a) Write a C program to sort the given numbers in ascending and descending order.

(b) Write a C program to add two matrices.

---------- Cut here ----------

5. (a) Write a C program to find the transpose of a matrix.

(b) Write a C program to remove repeated characters from a string.

---------- Cut here ----------

6. (a) Write a C program to process student records by using structures.

(b) Write a C program to convert uppercase letters into lower case and lower case letters into uppercase in a text file.

---------- Cut here ----------

7. (a) Write a C program to find the highest common factor of two positive integers.

(b) Write a program to convert the decimal number 55 into an equivalent binary number.

---------- Cut here ----------

8. (a) Write a C program to sort a set of names in an alphabetical order using pointer technique.

(b) Two data files Ladies and Gents contain sorted list of names of girls and boys in a class respectively. Write a C program to produce a third file 'students' which holds the merged list of the given files.

DISTANCE EDUCATION

M.Sc. (Software Engineering (5 Yrs Integrated) DEGREE EXAMINATION, MAY 2012.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a menu driven program in C to find an element using linear and binary search methods.

   (b) Write a C program to copy contents of one file to another file.

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2. (a) Write a C program to evaluate a mathematical expression using stack.

   (b) Write a C program to sort a set of elements using Insertion sort.

---------- Cut here ----------

3. (a) Write a C program to implement insert and delete operations on linked list structure.

   (b) Write a C program to copy contents of one file to another file.

4. (a) Write a C program to sort 10 numbers in ascending order with naming of variables and the value before and after sorting.

(b) Write a C program to implement PUSH and POP operations on stack.

---------- Cut here ----------

5. (a) Write a C program to implement insert and delete operations on queue using array concept.

(b) Write a C program to sort a set of elements using selection sort.

---------- Cut here ----------

6. (a) Write a C program to sort a set of elements using Insertion sort.

(b) Write a C program for linked list implementations of queue operations.

---------- Cut here ----------

7. (a) Write a C program to evaluate the given mathematical expression using stack.

(b) Write a C program to implement insert and delete operations on queue using array concept.

---------- Cut here ----------

8. (a) Write a C program to implement insert and delete operations on linked list structure.

(b) Write a C program to sort a set of elements using selection sort.

DISTANCE EDUCATION

M.Sc. (Software Engineering (5 Yrs. Integrated)) DEGREE EXAMINATION, MAY 2012.

C++ LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

Each subdivision carries 50 marks.

1. (a) Create a class called TIME that has integer data elements for hours, minutes, seconds. The constructors should initialize these data elements to specified value, if given and otherwise to O. A member function should display it in 11:50:45 format. The final member function should add two objects of type time passes an arguments.

   (b) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

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2. (a) Create a class that imitates part of the functionality of the basic data type int. Call the class int. The only data in this class is an integer variable. Include member functions to initialize an int. to zero, to initialize it to an integer value, to display it, and to add two int. values. Write a program that exercise this class by creating to initialized and one uninitialized int. values, adding these two initialized values and placing the sum in the uninitialized value and then displaying the result.

   (b) Using operator overloading, write a C++ program to find the difference and total length of two various tubes specified in meters and centimeters.

3. (a) Assume you want to generate a table of multiples of any number. Write a program that allows the user to enter the number and then generates the table, formatting it into ten columns and 20 lines.

(b) Raising a number n to power p is the same on multiplying by itself t times. Write a function called power () that takes a double value for n and int value for p, and returns the result as double. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main() function that gets values from the user to test this function.

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4. (a) Write a program to process students marks with the help of classes. All subjects mark must be ≥ 50 for pass otherwise fail.

(b) Create an equivalent of a four-function calculator. The program should request the user to enter a number, an operator and another number. It should then subtract and divide the two numbers. Finally it should display the result. When it finishes the calculation, the program should ask if the user wants to do another calculation.

5. (a) Using dynamic constructors write a C++ program to concatenate two given strings.

(b) Imagine a publishing company that markets both books and audio-cassette versions of its works. Create a class publication that stores the title (string) and price (type float) of a publication. From this class derive two classes.

Book, which adds a page count (type int) and TAPE, which adds a length count (type int). Each of these three classes should have a get data () function to get its data from the user at the keyboard and a putdata() function to display its data. Write a main() program to test the book and tape classes by creating instances of them, asking the user to fill in their data with getdata() and then displaying the data with putdata().

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6. (a) Create a class employee that contains employee number, employee name and address. Write a menu driven C++ program to get the 'n' number of employee details and display all details in employee namewise sorted order.

(b) Write a menu driven C++ program to add and subtract given two matrices of order m × n defined in class using operator overloading.

7. (a) Write a C++ program to prepare the payslips for 'n' number of employees using the following details.

D.A = 40% of basic pay

H.R.A. = 25% of basic pay

G-pay = Basic pay + DA + HRA

N-pay = G-pay – Deduction.

The result of program in the given format.

Emp. No.	Emp. Name	Basic pay	DA	HRA	LIC	PF	G-pay	N-pay

(b) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

---------- Cut here ----------

8. (a) Create a class of DONOR that contains donor number, donor name, age, address, sex, blood group. Write a program using C++ to print out the number, name and address of the donars for the following categories.

(i) Blood donors having O+ blood group

(ii) Blood donors in the age group between 16 to 25

(iii) Female donors with 'A' blood group in the age between 19 to 24

(b) Using pointes. Create a class and write a program to get n names and display them in sorted order.

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)DEGREE EXAMINATION, MAY 2012.

RELATIONAL DATABASE MANAGEMENT SYSTEMS — LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

1. Write a PLSQL program to prepare a mark statement of a student using student table and mark table. Assume your own data.

---------- Cut here ----------

2. Write a SQL program for the canteen maintenance with two tables namely menu and bill.

The menu table contains the following fields

Item, Unit-rate. The following items are available in the canteen;

Tea, Coffee and cool-drinks. The bill table contains the following fields.

EmpNo, Name, Date of issue, Item1, No of tokens issued for Item1, Rate1, Item2, No of tokens issued for item2, rate 2, item3 No of tokens issued for item 3, rate 3, total amount.

Note: An employee is allowed to get the token only once in a month.

Rate = Unit-Rate* No of tokens for particular item

Total Amount = Rate 1+Rate 2 + Rate 3

Prepare a report in the following format

S.No. Emp No. Name Purchased Rate

3. A newspaper vendor buys newspapers on wholesale from a distributor for Rs. 4 and sell them in retail for Rs. 4.50. At the end of the day, the onsold papers are returned to the distributor for Re. 1 rebate per paper. Write a program using Oracle to prepare a port for the newspaper vendor in the following format with 9 weeks date.

Week	Bought	Sold	Return	Profit	Loss
1	100	80	xx	xxxx	xxx
2	50	30	xx	xxxx	xxx
Total					

---------- Cut here ----------

4. A company Rx Pvt. Ltd. States monthly salary to its employee. It consists of basic salary, allowance, deduction

D.A. – 45% of Basic Salary.

H.R.A. – 7% of (Basic salary + DA)

Deductions

P.F. – Subscribed by an capital

L.I.C. Premium – Payable by an employee, the salary saving scheme.

Loan Recovery – if any payable by an employee.

Create a master table of 20 records with the following fields

Emp. No., P.F. Subscription, LIC premium, Loan recovery

Write a program in SQL to prepare a report with following information

Serial No., Emp. No., Bank AC/No, Name, Basic Salary, Total allowances, gross pay, Total deductions, net pay.

5. An airline reservation database contain the reservation table and personal table

Reservation table contains following fields

Flight No, Class, Passenger Name, Seat Number Personal table contains.

Passenger Name, Sex, age, martial status, Nationality.

Write a program to prepare the following reports :

(a) List of passengers names with seat number according to class and flight numbers.

(b) Total No of married female passengers in particular flight.

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6. Write a program in SQL for Hospital billing system with the following fields : Patient Name, No, Age, Attending Doctor 7% pe (In/out), consulting charge, Blood test, charge, X-ray charges, other test charges and total fee :

Prepare reports with the following conditions

(a) Patients undergone blood test

(b) Patients who have taken X-Rays

(c) Patients belong to In-patient category.

---------- Cut here ----------

7. A daily sales file contains record with the following fields. Department Number, date, item, description, sales price for each item, quantity, cost.

Write a program to list all input data and compute total amount of sales and profit for each sale. Accumulate and print total of sales and profit.

8. An examination has been conducted to a class of bo students and the following are given as inputs: Three test marks, Register No and Name.

Assign a letter grade to each student based on the average store and list register number, Average score, grade The minimum pass mark for each subject is 50. Write an SQL program for the above.

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